## MCS Saturday 7th-SHSAT

Instructor: Hsiao 'Hue' Weng · weng.hue.8@gmail.com

Grade 7 SHSAT Math · CAT Digital Format

# Multiplier Chain Control Lab

Session 4 Homework Activity Sheet · New Questions

**Name** _____ **Date** _____

**Completion Check**

Accuracy · model choice · CAT commit routine

| 18 Core | 8 Transfer | 26 Total |
|---|---|---|
| | | |

## A CAT Commit Protocol

5 prompts · 7 min

Practice the forward-only routine. Describe an observable action—not a vague promise to “be careful.”

**1 · Target** | **2 · Model** | **3 · Compute** | **4 · Audit** | **5 · Commit**

### A1 · Target first — Recall

A checkout problem mentions a list price, discount amount, tax, and later reward. What should you mark before calculating so that you answer the correct quantity?

Underline the exact quantity asked for and its units, such as today's checkout charge.

### A2 · Predict direction — Apply

A price is reduced 28% and then taxed 6%. Without calculating the exact result, state the comparison you expect between the final and original prices. Name the evidence.

Below the original: 6% tax on the reduced base is smaller than the 28% original-price discount.

### A3 · Entry discipline — Apply

Your work gives **$47.50**, but the typed-entry field accepts one number. State exactly what you enter and what you omit.

Enter 47.50. Omit the dollar sign, units, and any explanation.

### A4 · Productive audit — Reason

You used simple interest for 15 months and obtained more than two full years of interest. Name the first conversion you would inspect and explain why.

Check t = 15/12 = 1.25 years, not 15 years. At the same P and r, 1.25 years earns less than 2.

### A5 · Forward-only pacing — Defend

An unfamiliar item has consumed about 75 seconds. Write a two-sentence decision rule that protects both accuracy on this item and time for later items.

If I can name a workable model, I will take one short calculation step and check the target and format. If I am still stuck, I will eliminate impossible choices, enter my best supported answer, and commit rather than restart.

## B Model Translation Warm-Up

6 questions · 9 min

Write the multiplier, chain, or time conversion before doing any arithmetic.

**B1** Level 1

Write the multiplier for an **18% increase**.

1.18

One decimal

**B2** Level 1

Write the multiplier for a **32% decrease**.

0.68

One decimal

**B3** Level 2

A value rises 12% and then falls 15%. Write, but do not evaluate, the multiplier chain.

**Chain** 1.12 × 0.85

**B4** Level 2

Convert **21 months** to years for **I = Prt**.

21/12 = 7/4 = 1.75 years

Fraction or decimal

**B5** Level 2

Which expression models a **$14** fee added after a 25% discount on price **p**?

**A.** 0.75(p + 14)

**B.** 0.75p + 14

**C.** 1.25p + 14

**D.** p − 0.25(14)

B. 0.75p + 14

**B6** Level 3

Which single multiplier represents a 10% decrease followed by a 25% increase?

0.90 × 1.25 = 1.125

1.125

Evaluate the product

## C Core Accuracy Set

12 questions · 22 min

Treat each card as one forward-only screen: name the model, solve, scan the requested form, then check the commit box.

**Translate the percent · Track the current base · Convert months to years · Answer the named quantity**

**C1** **Level 1**

A quantity of 85 increases by 16%. What is the new quantity?

Increase: 85 × 1.16 = 98.6

98.6

☑ I used an increase multiplier.

**C2** **Level 1**

A membership count of 260 decreases by 35%. What count remains?

Decrease: 260 × 0.65 = 169

169

☑ I answered the amount remaining.

**C3** **Level 2**

A **$144** service is discounted 15% and then taxed 7.5%. What is the final charge?

Discount, then tax:
144 × 0.85 × 1.075 = 131.58

$131.58

☑ I rounded money to the nearest cent.

**C4** **Level 2**

A value of 550 increases 8% and then decreases 12%. Find the final value.

Successive changes:
550 × 1.08 × 0.88 = 522.72

522.72

☑ I multiplied successive factors.

**C5** **Level 2**

Find the simple interest on **$1,250** at 4.4% per year for 3 years.

Simple interest: I = Prt
1,250 × 0.044 × 3 = 165

$165

☑ I reported interest, not balance.

**C6** **Level 2**

How much simple interest does **$960** earn at 6.25% per year for 14 months?

t = 14/12 = 7/6 years
I = 960 × 0.0625 × (7/6) = 70

$70

☑ I converted months to years.

**C7** Level 3

After a 15% increase, a quantity equals 184. What was the original quantity?

Reverse increase:
184 ÷ 1.15 = 160

160

☑ I divided by the full multiplier.

**C8** Level 3

After a 25% discount and then 8% tax, an item costs **$121.50**. Find its original price.

Reverse the full chain:
121.50 ÷ (0.75 × 1.08) = 150

$150

☑ I reversed the entire chain.

**C9** Level 3

A price increases 60%. What percent discount from the increased price restores the original price?

1.60(1 - d) = 1
d = 1 - 1/1.60 = 0.375 = 37.5%

37.5

Enter the number without the percent sign.

**C10** Level 3

A deposit earns **$126** simple interest at 7% per year for 18 months. Find the principal.

t = 18/12 = 1.5 years
P = 126 ÷ (0.07 × 1.5) = 1,200

$1,200

☑ I solved for **P**, not total balance.

**C11** Level 3

**$900** earns **$99** simple interest in 2.5 years. What annual interest rate was used?

r = 99 ÷ (900 × 2.5)
r = 0.044 = 4.4%

4.4

Enter the numerical percent.

**C12** Level 4

A borrower receives **$1,800** at 5.2% simple interest for 30 months. What total amount is owed?

t = 30/12 = 2.5 years
I = 1,800 × 0.052 × 2.5 = 234
Total = 1,800 + 234 = 2,034

$2,034

☑ I added principal to interest.

# D SHSAT Transfer Set

8 questions · 20 min

Model the timeline before calculating. Separate percentage changes, fixed charges, and later credits.

### D1 · Checkout timeline

Level 3

**Appliance Pickup Notice.** A **$240** appliance is discounted 22%. Sales tax of 8.25% is applied to the discounted price. An **$18** delivery fee is then added and is not taxed. A **$12** reward for a future purchase does not reduce today's charge. What is charged today?

Discounted price = 240 × 0.78 = 187.20
Taxed price = 187.20 × 1.0825 = 202.644
Today: 202.644 + 18 = 220.644
The future reward is not subtracted.

$220.64

Round to the nearest cent.

### D2 · Compare order

Level 3

A **$160** item has two offers. Store A takes 25% off, then subtracts an **$18** coupon. Store B subtracts **$18**, then takes 25% off. Which store is lower, and by how much?

Store A: 160 × 0.75 - 18 = $102.00
Store B: (160 - 18) × 0.75 = $106.50
Difference: 106.50 - 102 = $4.50
Store A is lower by $4.50.

### D3 · Net effect

Level 3

A store raises a price 30% and later discounts the new price 20%. Relative to the original, what is the overall percent change?

**A.** 10% decrease
**B.** 4% increase
**C.** 6% increase
**D.** No change

1.30 × 0.80 = 1.04
B. 4% increase

### D4 · Reverse balance

Level 4

An account's balance is **$2,192** after earning 4.8% simple interest for 2 years. What was the principal?

Balance = P(1 + rt)
2,192 = P(1 + 0.048 × 2)
P = 2,192 ÷ 1.096 = 2,000

$2,000

The balance is principal plus interest.

### D5 · Bonus rule

Level 4

A **$3,200** fund earns 3.75% simple interest for 28 months. A sponsor then adds a bonus equal to 15% of the interest earned. Find the combined interest and bonus.

t = 28/12 = 7/3 years
I = 3,200 × 0.0375 × (7/3) = $280
Bonus = 280 × 0.15 = $42
Combined = 280 + 42 = $322

$322

### D6 · Subscription notice

Level 4

**Renewal Order.** An annual plan costs **$480**. At renewal, the price increases 12%. A **$35** loyalty credit is then subtracted. Tax of 9% is applied to the amount after the credit. What is the final renewal charge?

Increased price = 480 × 1.12 = 537.60
After credit = 537.60 - 35 = 502.60
Taxed total = 502.60 × 1.09 = 547.834

$547.83

Round to the nearest cent.

### D7 · Multi-select

Level 4

Select **all** expressions equivalent to decreasing **x** by 10% and then increasing the result by 25%.

**A.** 0.90(1.25x)

**B.** 1.125x

**C.** x + 0.15x

**D.** x + 0.125x

A, B, and D
0.90 × 1.25x = 1.125x = x + 0.125x.
Choice C equals 1.15x, not 1.125x.

### D8 · Distractor analysis

Level 4

A student says, “A 20% discount followed by a 20% markup returns any price to its start.” Identify the exact misconception and use an original price of **$150** to disprove the claim.

The student treats both percentages as if they use the same base. The markup applies to the discounted price, not the original.
$150 × 0.80 = $120
$120 × 1.20 = $144
The $24 markup does not undo the $30 discount; the result is $6 below the start.

**Correct final price** $144

## E One-Screen Rehearsal

3 decisions · 5 min

Imagine that advancing permanently closes the item. State the last useful check—not a full re-solve.

**E1.** Two multipliers produce **1.02**. The original price was **$300**, but your answer is below **$300**. What contradiction should stop your commit?

A multiplier of 1.02 means a 2% increase, so the result must exceed $300, not fall below it.

**E2.** You found **$84** of interest, but the final sentence asks for the account balance. What final action is required?

Add the $84 interest to the original principal: balance = principal + $84.

**E3.** Your rate equation gives **0.045**, while the response asks for the numerical value of the percent. What belongs in the field?

4.5 (convert 0.045 to 4.5%; omit the percent sign).

## F Process Debrief

Evidence, not impressions

Complete this before checking solutions. Focus on decisions that can be repeated under forward-only CAT conditions.

- ☐ I marked the exact quantity requested before calculating.
- ☐ I applied each percent to the value that existed at that step.
- ☐ I converted months to a fraction of a year in simple interest.
- ☐ I reversed a chain by dividing by the complete product.
- ☐ I converted each percent change to a multiplier.
- ☐ I kept fixed fees and later credits separate from percent changes.
- ☐ I distinguished interest from total balance.
- ☐ I entered only the value and form requested.

## Timing Record

Find the pace change

| | | |
|---|---|---|
| **A–B** | _____ | min |
| **C** | _____ | min |
| **D–E** | _____ | min |
| **Total** | _____ | min |

**Pace evidence.** Where did your pace change? Name the model or reading demand that caused it.

## G Error Log

Up to 5 corrections

Choose one primary type: Model, Procedure, Reading, Arithmetic, Format, Wrong Quantity, or Pace.

| Problem | First wrong decision | Primary type | Next-time interruption cue |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

## H Transfer Reflection

Complete sentences

**1. The wording clue that most reliably tells me which percent model to use is**

The clues "increases by" and "decreases by" tell me to use 1 + r and 1 - r; "then" tells me to multiply successive factors.

**2. One changing-base error I can now detect before calculating is**

I can detect the mistake of adding or subtracting successive percentages as if both apply to the original base.

**3. When months appear in a simple-interest problem, my next observable action will be**

I will write t = number of months / 12 in years before substituting into I = Prt.

**4. My five-second commit question for a typed-entry item will be**

I will ask, "Does my entry answer the exact quantity requested, in the required form and with a reasonable size?"

Bring the completed activity sheet and error log to the next session.